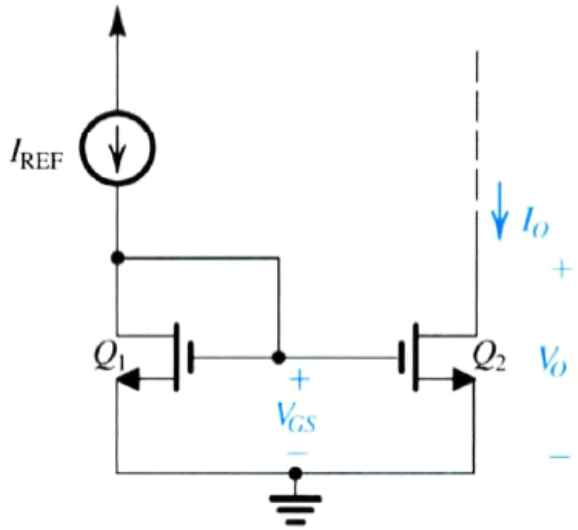
Reading

Adel S. Sedra and Kenneth C. (KC) Smith, Tony Chan Carusone, and Vincent Gaudet
Microelectronic Circuits 8th Edition, *Oxford University Press*, 2019.

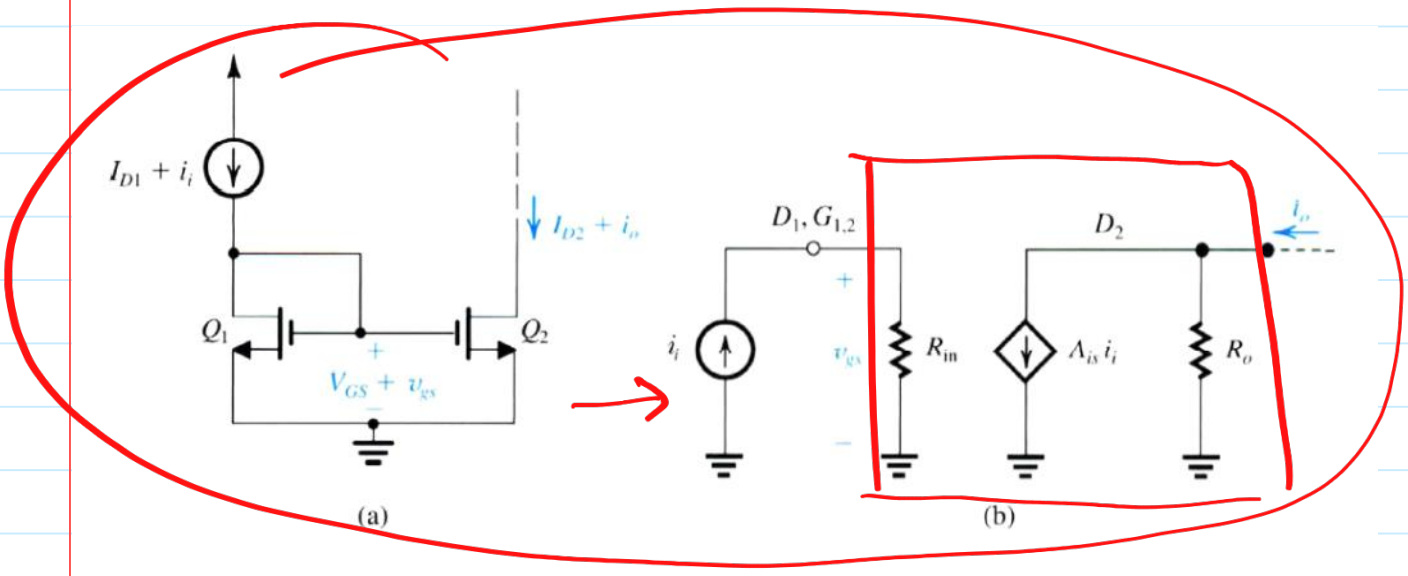
- Chapter 9.5 Differential Amplifier with a Current-Mirror Load
- Chapter 9.6 Multistage Amplifiers

Recap



$$I_O|_{\text{nominal}} = \frac{(W/L)_2}{(W/L)_1} I_{REF}$$

$$I_O = \frac{(W/L)_2}{(W/L)_1} I_{REF}\left[1 + \frac{V_O - V_{GS}}{V_{A2}}\right]$$

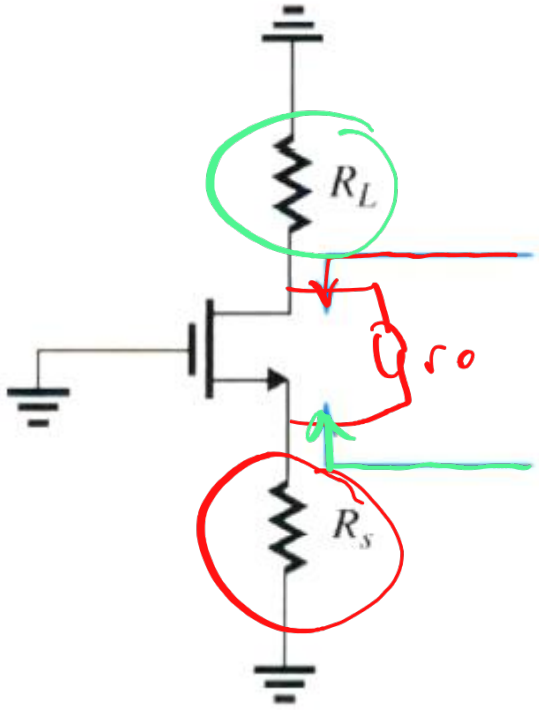


$$A_{is} = \frac{(W/L)_2}{(W/L)_1}\left[\frac{1}{1 + (1/g_{m1}r_{o1})}\right]$$

$$R_{in} = r_{o1} \| \frac{1}{g_{m1}} \simeq \frac{1}{g_{m1}}$$

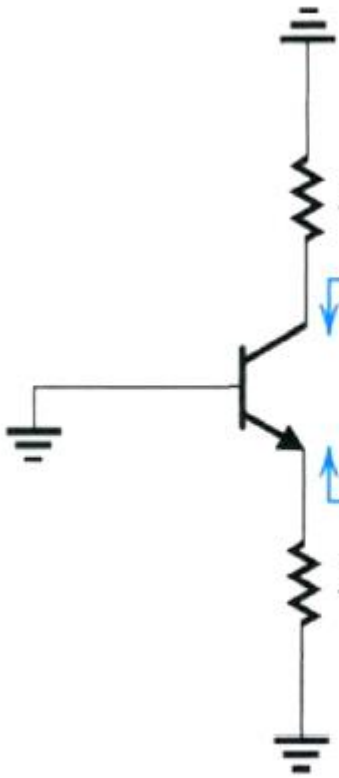
$$R_o = r_{o2}$$

Recap



$$R_{out} = r_o + R_s + g_m r_o R_s$$
$$\simeq r_o + (g_m r_o) R_s$$

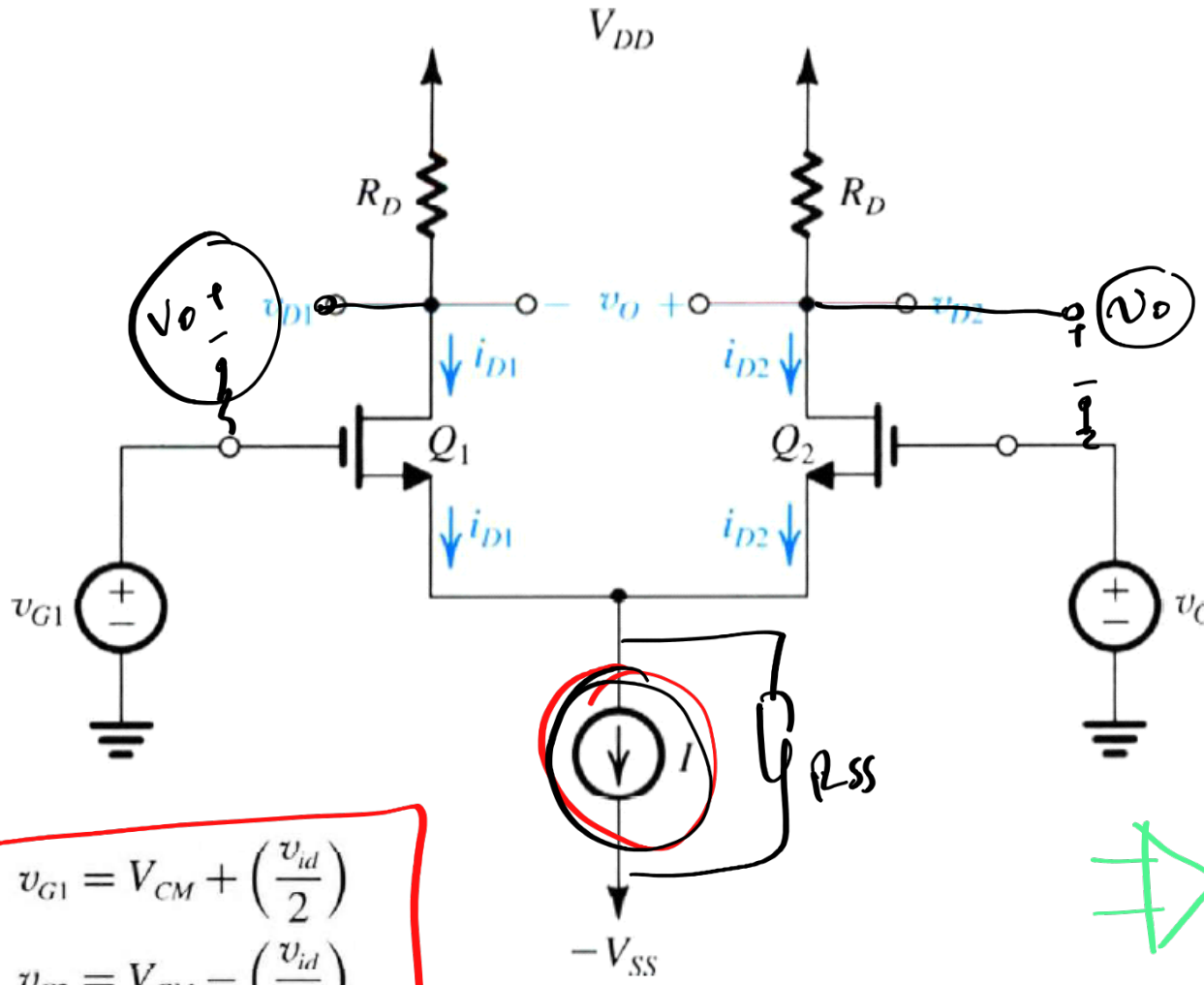
$$R_{in} = \frac{r_o + R_L}{g_m r_o + 1}$$
$$\simeq \frac{1}{g_m} + \frac{R_L}{g_m r_o}$$



$$R_{out} = r_o + (R_e \| r_\pi) + g_m r_o (R_e \| r_\pi)$$
$$\simeq r_o + (g_m r_o)(R_e \| r_\pi)$$

$$R_{in} = r_\pi \left\| \left[\frac{r_o + R_L}{g_m r_o + 1} \right] \right.$$
$$\simeq \frac{1}{g_m} + \frac{R_L}{g_m r_o}$$

Recap



$$v_{G1} = V_{CM} + \left(\frac{v_{id}}{2}\right)$$
$$v_{G2} = V_{CM} - \left(\frac{v_{id}}{2}\right)$$

$$A_d \equiv \frac{v_{od}}{v_{id}} = \frac{v_{o2} - v_{o1}}{v_{id}} = g_m R_D$$

$$A_d \equiv \frac{v_{od}}{v_{id}} = g_{m1}(r_{o1} \parallel r_{o3})$$

$$A_{cm} = -\left(\frac{R_D}{2R_{SS}}\right)\left(\frac{\Delta R_D}{R_D}\right)$$

$$A_{cm} \simeq \left(\frac{R_D}{2R_{SS}}\right)\left(\frac{\Delta g_m}{g_m}\right)$$

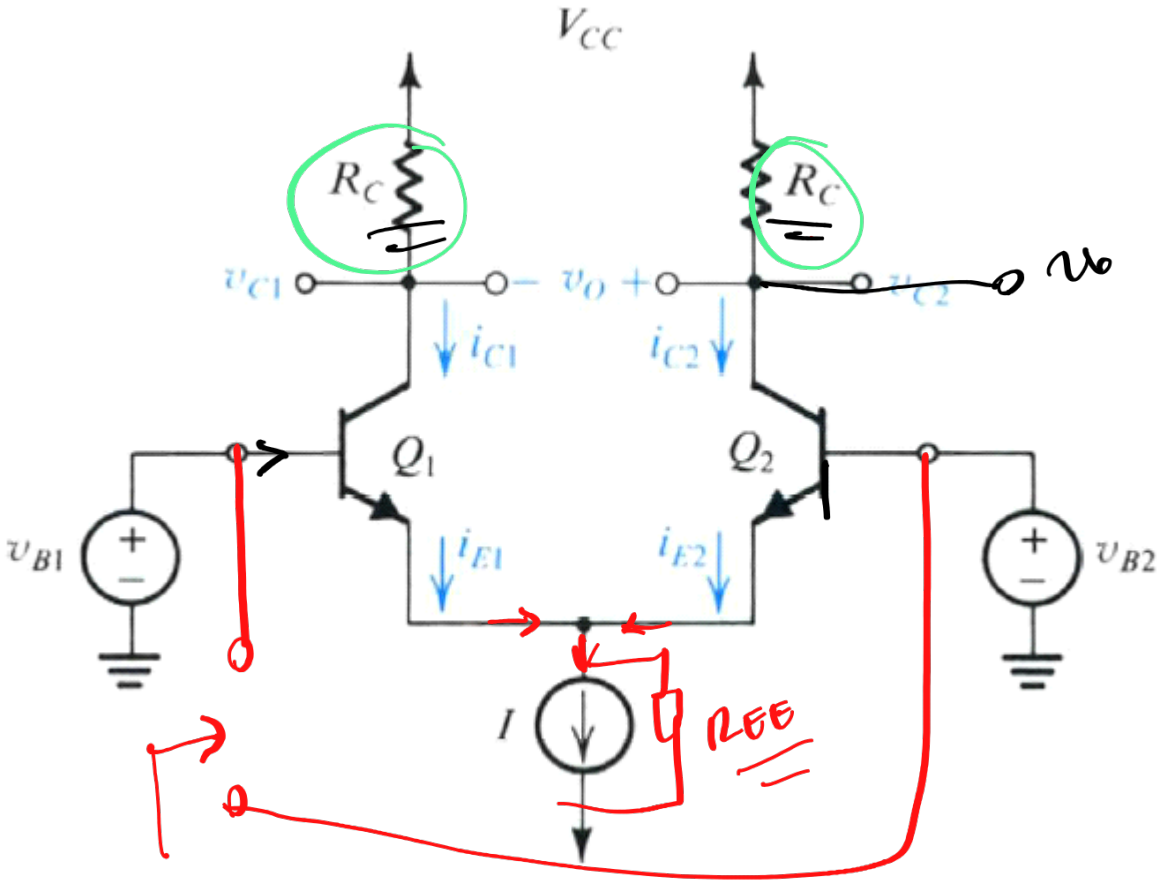
$$\text{CMRR} \equiv \frac{|A_d|}{|A_{cm}|}$$

$$A_d = -\frac{1}{2} g_m R_D$$

$$A_{cm} = -\frac{R_D}{2R_{SS}}$$

$$\text{CMRR} = g_m R_{SS}$$

Recap



$$R_{id} \equiv \frac{v_{id}}{i_b} = (\beta + 1)2r_e = 2r_\pi$$

$$R_{id} = (\beta + 1)(2r_e + 2R_e)$$

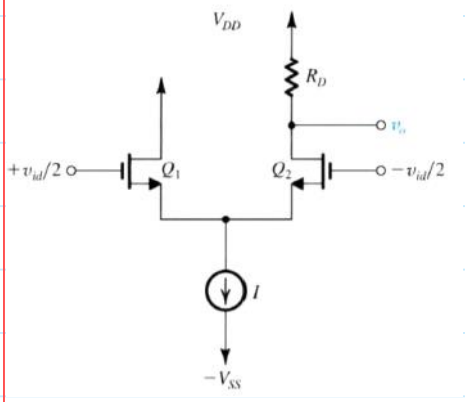
$$A_d = \frac{v_{od}}{v_{id}} = g_m R_C = g_m (r_o \| r_o)$$

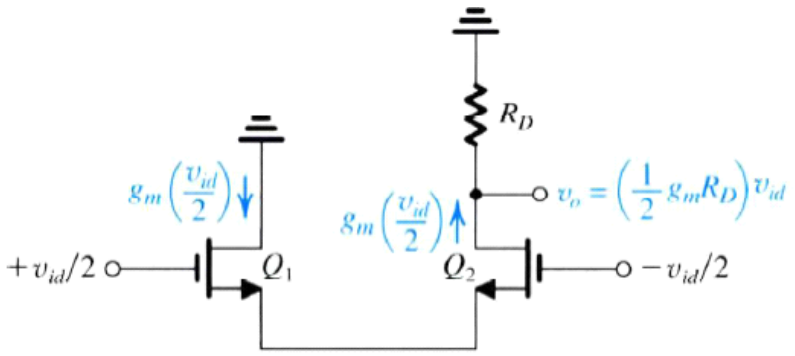
$$A_d = \frac{\alpha(2R_C)}{2r_e + 2R_e} \simeq \frac{R_C}{r_e + R_e}$$

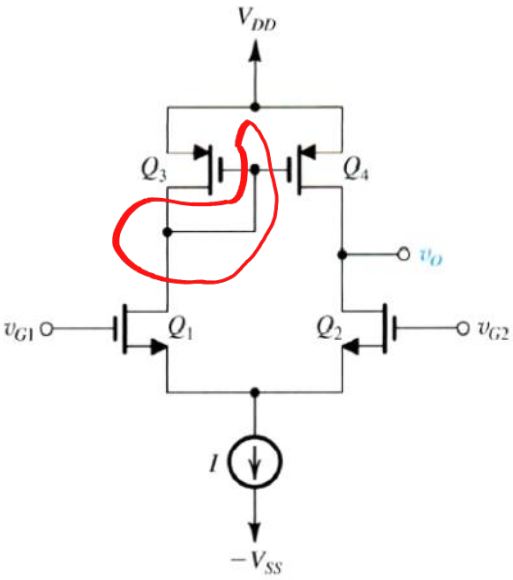
$$A_{cm} \simeq -\left(\frac{R_C}{2R_{EE}}\right)\left(\frac{\Delta R_C}{R_C}\right)$$

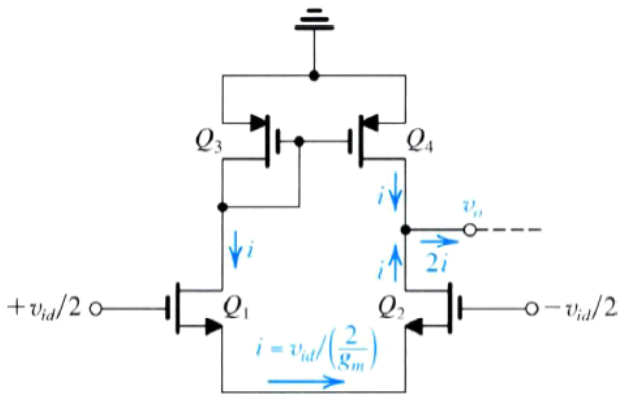
$$R_{icm} \simeq \beta R_{EE} \frac{1 + R_C/\beta r_o}{1 + \dfrac{R_C + 2R_{EE}}{r_o}}$$

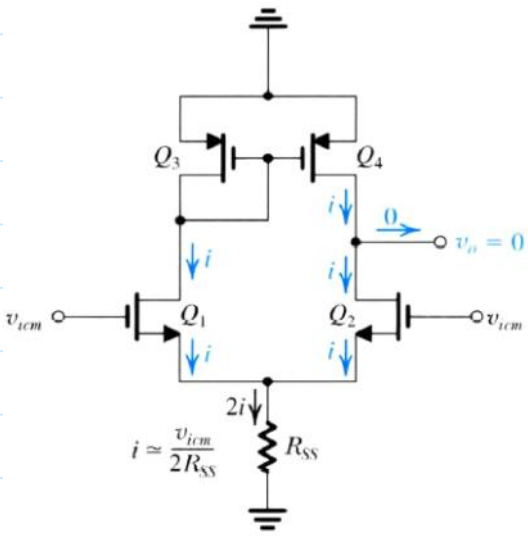
The Current Mirror Loaded MOS DiffAmp



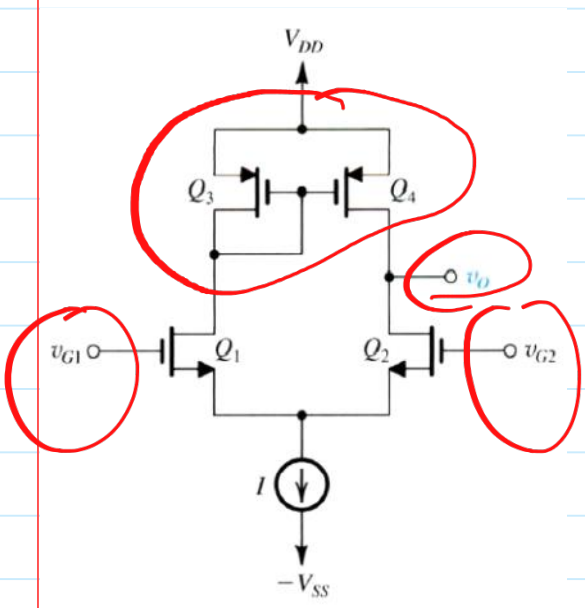


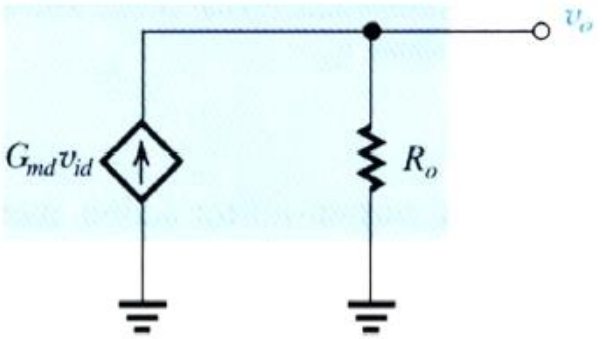






Differential Gain of the MOS Pair



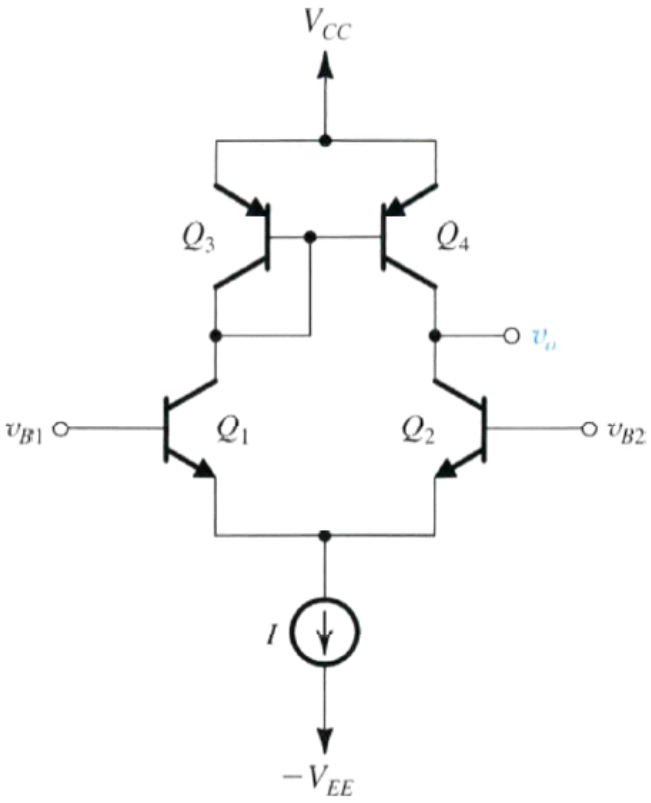


$A_d = \frac{v_o}{v_{id}} = G_{md}$. R_o

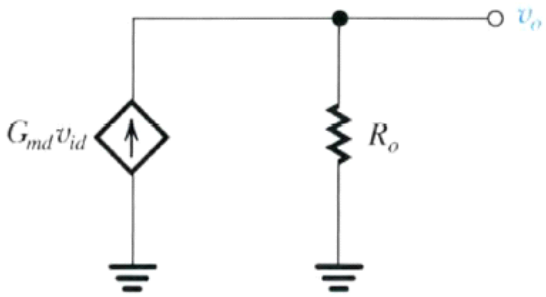
$$G_{md} = g_{m1,2} \qquad R_o = R_{od} \parallel R_{om} \qquad R_o = r_{o2} \parallel r_{o4}$$

$$A_d \equiv \frac{v_o}{v_{id}} = G_{md}R_o = g_{m1,2}(r_{o2} \parallel r_{o4})$$

Differential Gain of the Bipolar Pair



(a)



(b)

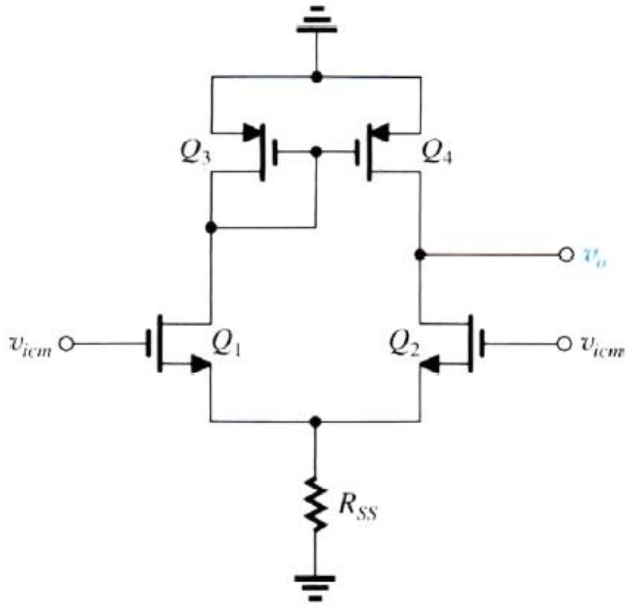
$$G_{md} = g_{m1,2}$$

$$R_o = r_{o2} \parallel r_{o4} = R_{od} \parallel R_{om}$$

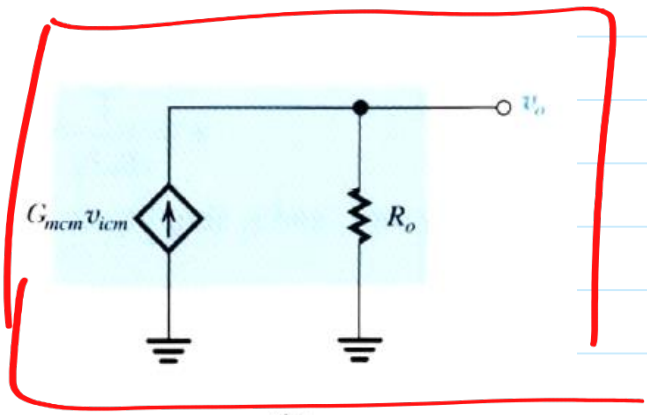
$$A_d \equiv \frac{v_o}{v_{id}} = G_{md}R_o = g_m(r_{o2} \parallel r_{o4})$$

$$R_{id} = 2r_\pi$$

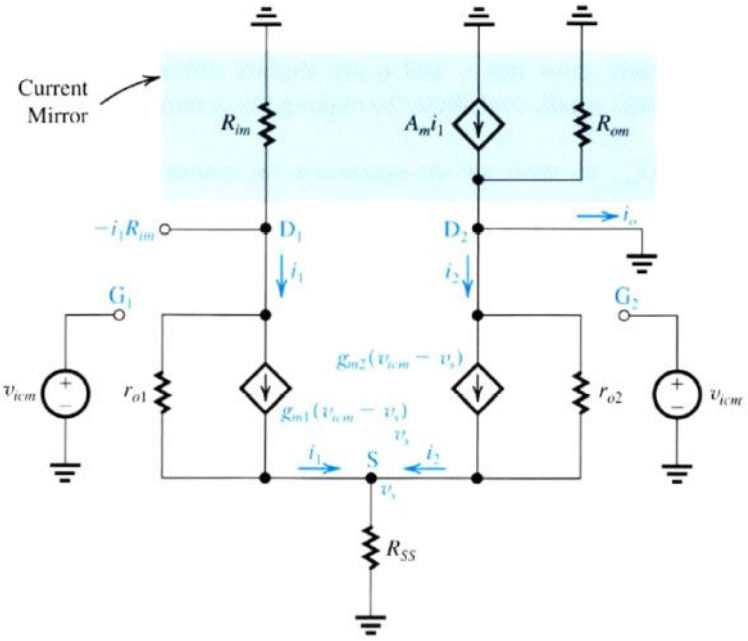
Common-Mode Gain of the MOS Pair



(a)



(b)



$$i_1 \simeq \frac{v_{icm}}{2R_{SS}}\left(1 - \epsilon - \frac{R_{im}}{2r_o}\right)$$

$$i_2 \simeq \left(\frac{v_{icm}}{2R_{SS}}\right)\left(1 - \epsilon + \frac{R_{im}}{2r_o}\right)$$

$$\epsilon = \frac{1}{g_m r_o} + \frac{1}{2g_m R_{SS}} \ll 1$$

$$G_{mcm} \simeq -\frac{1}{2R_{SS}} \frac{1}{g_{m3}(r_o \parallel r_{o3})}$$

$$A_{cm} = -\frac{1}{2g_{m3}R_{SS}}$$

$$\text{CMRR} = (2g_{m1,2}R_{SS})[g_{m3}(r_o \parallel r_{o3})]$$

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